SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING) Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)

Course Code:	ITA0609	**Course Title:**	Machine Learning
CO Assessed:	CO1 – Analyze (BL 3)	**Assessment:**	Assessment Tool 1 – Application Based Questions.
Weightage:	40% of CIA	**Total Marks:**	30 (3 Questions ×10 Mark)
CO1 Statement:	*To acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search (BL1)*		
Student Name:	G. Venkata Sai Sudha	**Reg. No.:**	192412397
Section / Batch:	SLOT B	**Date:**	15.7.26

Code of Conduct

I (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Instructions.

- This test contains 5 Short questions, each carrying 10 mark.
- All questions are application-based and require analytical thinking and practical implementation. · No negative marking. Unanswered questions carry zero marks.
- Provide suitable algorithms, logic, calculations, or case-based solutions wherever required.

Annexure A – Application-Based Questions

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?(5 Marks)

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. **(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4**
2. **(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2**
3. **(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0**
4. **(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1**
5. **(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3**
6. **(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6**
7. **(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9**
8. **(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2**
9. **i) Identify the epoch at which the model begins to overfit.**
10. **ii) Suggest two effective techniques to reduce overfitting in this model.**
11. **iii) Discuss how overfitting affects the model's generalization ability on unseen data.**

3. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. **(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%**

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%
20. i) Identify the epoch at which the model starts overfitting.
21. ii) Explain why validation accuracy decreases despite training accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?(5 Marks)

Answer:

Section	**Red**	**Blue**	**Black**	**Total**
A	5	3	2	10
B	4	6	5	15
C	7	2	6	15

Since Section B is twice as likely to be selected:

- $P(A) = \frac{1}{4}$
- $P(B) = \frac{1}{2}$
- $P(C) = \frac{1}{4}$

The probabilities of selecting a blue car from each section are:

- $P(\text{Blue} \mid A) = \frac{3}{10}$
- $P(\text{Blue} \mid B) = \frac{6}{15} = \frac{2}{5}$
- $P(\text{Blue} \mid C) = \frac{2}{15}$

Step 1: Find $P(\text{Blue})$

Using the Total Probability Theorem,

- $P(\text{Blue}) = P(A)P(\text{Blue} \mid A) + P(B)P(\text{Blue} \mid B) + P(C)P(\text{Blue} \mid C)$

Substituting the values,

- $P(\text{Blue}) = \left(\frac{1}{4} \times \frac{3}{10}\right) + \left(\frac{1}{2} \times \frac{2}{5}\right) + \left(\frac{1}{4} \times \frac{2}{15}\right)$
- $P(\text{Blue}) = \frac{3}{40} + \frac{1}{5} + \frac{1}{30}$

Taking the LCM (120),

- $P(\text{Blue}) = \frac{9+24+4}{120} = \frac{37}{120}$

Step 2: Apply Bayes' Theorem

We need to find $P(C \mid \text{Blue})$.

- $P(C \mid \text{Blue}) = \frac{P(C)\,P(\text{Blue}|C)}{P(\text{Blue})}$

Substituting the values,

- $P(C \mid \text{Blue}) = \frac{\left(\frac{1}{4}\times\frac{2}{15}\right)}{\frac{37}{120}}$
- $P(C \mid \text{Blue}) = \frac{1}{30} \times \frac{120}{37}$
- $P(C \mid \text{Blue}) = \frac{4}{37}$

Final Answer

- $\boxed{P(C \mid \text{Blue}) = \frac{4}{37} \approx 0.1081}$

Therefore, the probability that the selected blue car came from Section C is approximately 10.81%.

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4

2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2

3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0

4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1

5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3

6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6

7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9

8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

9. i) Identify the epoch at which the model begins to overfit.

10. ii) Suggest two effective techniques to reduce overfitting in this model.

11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Answer:

WHAT IS THE PROBLEM?

To examine the performance of a machine-learning model using the given training loss and validation loss values over multiple epochs, identify when overfitting begins, suggest methods to reduce overfitting, and explain its effect on the model's performance on unseen data.

STEPS USED TO SOLVE THE PROBLEM

1. Observe the training loss and validation loss values for each epoch.
2. Compare how both loss values change across epochs.
3. Identify the point where training loss decreases while validation loss starts increasing.
4. Determine the epoch at which overfitting begins.
5. Suggest suitable techniques to reduce overfitting.
6. Discuss how overfitting affects the model's ability to generalize to unseen data.

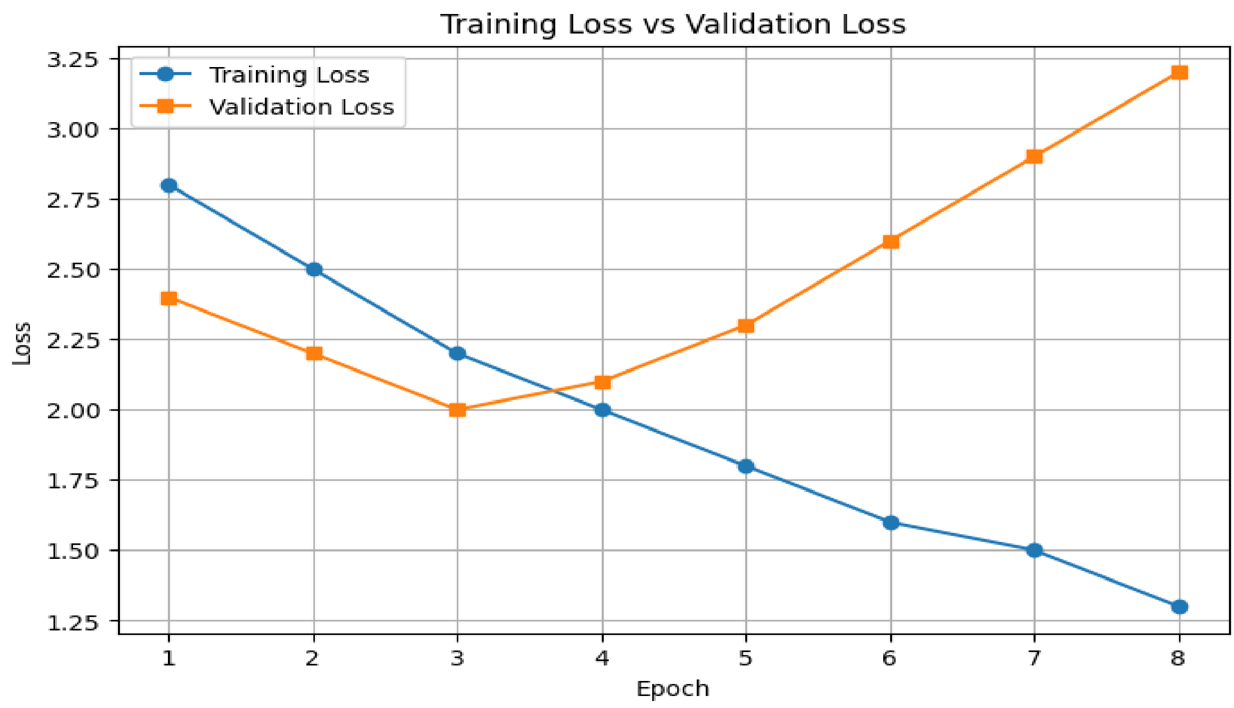
INPUT

Epoch-wise loss values:

- Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
- Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
- Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
- Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
- Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
- Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
- Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
- Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

OUTPUT

1. The model begins to overfit at Epoch 4.
2. Two techniques to reduce overfitting are:
    - Early Stopping
    - Regularization (Dropout or L1/L2 Regularization)
3. Overfitting reduces the model's generalization ability, causing poor performance on unseen data despite achieving low training loss.



3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

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18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.

21. ii) Explain why validation accuracy decreases despite training accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

Answer:

WHAT IS THE PROBLEM?

To examine the performance of a machine-learning model using the given training accuracy and validation accuracy values over multiple epochs, identify when overfitting begins, explain why validation accuracy decreases while training accuracy increases, and suggest methods to improve validation performance.

STEPS USED TO SOLVE THE PROBLEM

1. Observe the training accuracy and validation accuracy values for each epoch.
2. Compare how both accuracy values change across epochs.
3. Identify the point where training accuracy continues to increase while validation accuracy starts decreasing.
4. Determine the epoch at which overfitting begins.
5. Explain why validation accuracy decreases despite improvements in training accuracy.
6. Suggest suitable methods to improve validation performance.

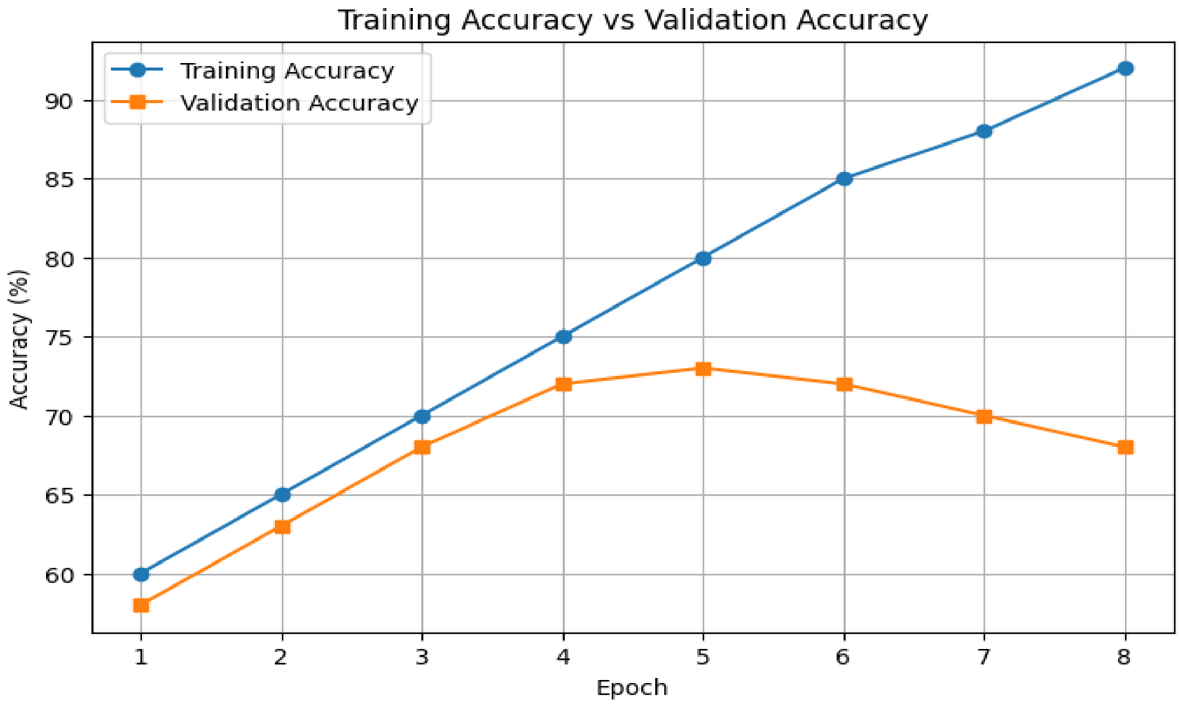
INPUT

Epoch-wise accuracy values:

- Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
- Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
- Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
- Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
- Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
- Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
- Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
- Epoch 8: Training Accuracy = 92%, Validation Accuracy = 68%

OUTPUT

1. The model starts overfitting at Epoch 6.
2. Validation accuracy decreases because the model begins to memorize the training data instead of learning generalized patterns.
3. Two methods to improve validation performance are:
    - Early Stopping.
    - Regularization techniques such as Dropout or L2 Regularization.


Training Accuracy vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy (%)
90
85
80
75
70
65
60
1
2
3
4
5
6
7
8
Epoch